

MULTIMODAL AI FRAMEWORK FOR EARLY PARKINSON'S DISEASE DETECTION

25-26J-109

Project Proposal Report

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Science

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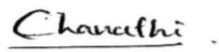
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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Signature of the supervisor:



Date: 2025-08-31

ABSTRACT

Parkinson's disease (PD) is a progressive neurodegenerative disorder primarily associated with motor symptoms, but cognitive impairments, particularly in executive, attentional, and visuospatial domains often emerge in the early stages. These subtle changes can precede significant disability, underscoring the importance of timely detection. Traditional diagnostic methods rely heavily on clinical evaluation, which can be subjective and limited in identifying early non-motor indicators. This research proposes a multimodal artificial intelligence (AI) framework that integrates neuropsychological and behavioral signals to predict early cognitive impairment in PD, with emphasis on enhancing explainability for clinical trust and adoption.

The methodology involves training predictive models using publicly available datasets comprising cognitive task performance, clinical scores, and neuropsychological indicators. Features are preprocessed, extracted, and integrated through multimodal learning techniques to improve prediction accuracy. The final system incorporates Explainable AI (XAI) mechanisms, enabling transparent interpretation of model outputs by highlighting feature importance and decision pathways. A mobile application prototype will be developed to operationalize the model, allowing patients to complete cognitive and behavioral tasks remotely. User task data will be processed in real time, analyzed by the trained model, and used to generate predictive insights regarding potential cognitive decline.

Anticipated results include improved sensitivity and specificity in detecting early cognitive impairment compared to conventional approaches, alongside interpretable outputs that foster clinical confidence. The expected contribution of this research lies in developing a clinically relevant, technologically feasible, and patient-centered solution that bridges the gap between AI-driven prediction and real-world healthcare integration. Findings may inform early intervention strategies, reduce diagnostic delays, and enhance quality of life for individuals at risk of Parkinson's-related cognitive decline.

Keywords: Parkinson's disease, Cognitive impairment, Multimodal learning, Explainable AI, Mobile health

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
DBS	Deep Brain Stimulation
LIME	Local Interpretable Model-Agnostic Explanations
ML	Machine Learning
MoCA	Montreal Cognitive Assessment
NLP	Natural Language Processing
PD	Parkinson's Disease
PPMI	Parkinson's Progression Markers Initiative
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Oversampling Technique
SVM	Support Vector Machine
TMT	Trail Making Test
TMT-B	Trail Making Test Part B
XAI	Explainable Artificial Intelligence

1.INTRODUCTION

1.1 Background and literature survey

Parkinson's disease (PD) is a progressive neurodegenerative disorder primarily characterized by motor symptoms such as tremors, rigidity, and bradykinesia. However, research increasingly emphasizes that non-motor symptoms, particularly cognitive impairment, are also common and often emerge in the early stages of the disease [1]. Cognitive changes in PD typically affect executive function, attention, visuospatial ability, and processing speed, and these deficits can precede severe motor disability, making them highly relevant for early diagnosis and monitoring [2].

The rapid growth of **digital biomarkers** has introduced new possibilities for remote, continuous, and scalable measurement of PD-related symptoms. Mobile applications and wearable devices have shown strong potential in capturing subtle disease-related changes, enabling longitudinal tracking outside traditional clinical settings [3]. Nevertheless, most digital biomarker implementations to date have emphasized motor functions, while cognitive impairments, though equally significant remain less explored in digital contexts [4].

Large-scale initiatives such as the mPower study (2016) have demonstrated the feasibility of collecting smartphone-based data from thousands of participants, including tasks such as tapping, reaction time, and memory matching [5]. Similarly, the PD Digital Biomarker DREAM Challenge highlighted the potential of machine learning to predict disease states using mobile task data [6]. These studies confirm the value of mobile-based data collection, but analyses have primarily focused on motor outcomes, with limited use of fine-grained cognitive features.

Classical neuropsychological tasks continue to provide valuable insights into cognitive dysfunction in PD. For instance, the Trail Making Test (TMT), especially Part B, is widely used to assess executive control and has been shown to relate strongly to functional outcomes in PD patients [7]. The Stroop task provides a measure of response inhibition and processing speed and has been applied in both longitudinal PD

studies and peri-deep brain stimulation (DBS) contexts [8]. Recently, digital variants of these tasks have been piloted, demonstrating discriminatory potential in distinguishing PD patients from healthy controls [9]. However, most studies still rely on aggregate performance metrics (e.g., total time, total errors), which may overlook meaningful trial-level behavioral patterns.

Emerging literature suggests that fine-grained cognitive features such as trial-to-trial variability in reaction times, switching costs, commission/omission micro-errors, and fatigue or learning effects, can be more sensitive to early cognitive decline than mean scores alone [10]. Such micro-level metrics provide richer insight into cognitive processes, potentially enabling earlier detection of subtle impairments.

Broad screening tools such as the Montreal Cognitive Assessment (MoCA) remain widely used and have shown correlations with motor severity in PD [11]. While helpful as global screening instruments, these tools lack the precision and interpretability required for detailed analysis of early cognitive decline. This underscores the need for **digital cognitive biomarkers** that can capture subtle, fine-grained changes, while also providing interpretable outputs that align with clinical reasoning.

In this research, the focus is on **Cognitive Biomarker Analysis** through digital neuropsychological tasks, which will be integrated into the larger multi-modal mobile application developed by the group project. The app will combine inputs from speech, motion, drawing, and cognitive tasks into a single predictive system. This integration enables not only sensitivity in detecting early PD-related changes but also scalability and accessibility for real-world use.

1.2 Research Gap

Although cognitive impairment is well recognized as an early feature of Parkinson's disease, existing digital assessment approaches remain limited in scope and depth. A large proportion of studies rely on aggregate performance scores such as total time to complete a task or total number of errors [7], [8]. While these measures are useful for broad screening, they often fail to capture the subtle trial-to-trial fluctuations that may

provide earlier and more sensitive indicators of cognitive decline. Consequently, potentially valuable micro-level information such as reaction-time variability, switching costs, or commission and omission micro-errors is left unexplored.

Furthermore, although machine learning models have been increasingly applied in the context of digital biomarkers, their application to cognitive data in PD has largely been black box in nature. Many models emphasize accuracy but fail to provide interpretable outputs that clinicians can trust and act upon [3], [10]. Without explainability, adoption in clinical settings is severely limited, as health professionals require clear evidence of which cognitive control processes are contributing to predictions.

In addition, while large-scale public datasets such as the mPower study provide a unique opportunity to validate digital biomarkers at scale, their cognitive task data remain underutilized. Most prior analyses on such datasets have focused on motor signals, with only limited efforts to map cognitive signals (e.g., reaction time, memory matching) to established neuropsychological constructs such as executive control or attention [5], [6].

Finally, most prior research has examined **biomarkers in isolation**, either focusing on motor, speech, or cognitive modalities individually. There is a lack of work that integrates cognitive biomarkers into a **multi-modal system** alongside speech, motion, and drawing features within a unified mobile application. Such integration could provide a richer and more reliable detection framework, while also enhancing scalability for real-world use.

Table 1.1 provides a comparative summary of selected past studies in cognitive biomarker research and highlights the gaps addressed by the proposed study.

Table 1.1 Comparison of research

Study / Source	Cognitive Domain	Trial-Level Features	Explainability	Integration / Scope
TMT in PD [7]	Executive function	✗	✗	Aggregate-only metrics
Stroop in PD [8]	Inhibition, speed	✗	✗	Aggregate-only metrics
Digital TMT pilot [9]	Executive, visuospatial	✗	✗	Limited sample size
mPower dataset [5], [6]	Motor + cognition	✗	✗	Cognitive features underexplored
Trial-level variability study [10]	Processing speed	✗	✗	Limited validation and scope
Proposed Study	Executive function, attention, learning trends	✓	✓	Integrates cognitive biomarkers into a multi-modal system for early PD detection; scalable and interpretable

1.3 Research Problem

Parkinson’s disease (PD) is a progressive neurodegenerative disorder characterized by motor symptoms, but cognitive impairments often appear early and can serve as important indicators for timely diagnosis. Traditional clinical assessments and paper-based neuropsychological tests provide aggregate measures such as total task time or overall errors, which fail to capture subtle cognitive changes in domains like executive function, attention, memory, and learning. As a result, early cognitive decline often goes undetected until more overt symptoms appear.

While digital neuropsychological assessments and machine learning approaches have been explored, existing solutions are limited in interpretability and rarely integrate cognitive biomarkers with other modalities such as speech, motion, or drawing. This lack of integration reduces the potential accuracy and reliability of early PD detection. Moreover, most publicly available cognitive datasets remain underutilized, focusing primarily on motor features and ignoring fine-grained cognitive signals.

There is therefore a clear need for a system that can extract relevant cognitive features from neuropsychological tasks, interpret their contributions to PD prediction, and integrate them into a multi-modal mobile framework. Such a system would enable early, scalable, and clinically meaningful detection of Parkinson's disease, while providing transparent outputs that clinicians can trust.

2. OBJECTIVES

2.1 Main Objectives

To develop an interpretable cognitive biomarker analysis module that detects early signs of Parkinson's disease through neuropsychological and cognitive tasks, contributing to a multi-modal framework for the early detection of Parkinson disease.

2.2 Specific Objectives

- **To extract cognitive biomarkers from neuropsychological tasks**
Derive fine-grained cognitive and neuropsychological features from publicly available datasets that reflect executive functions, memory, attention, and processing speed. Metrics such as reaction-time variability, task-switching costs, error patterns, and learning/fatigue trends will be captured. These features are intended to provide sensitive indicators of Parkinson's-related cognitive decline, supporting predictive modeling and clinically meaningful interpretation.
- **To map extracted features to neuropsychological constructs for interpretability**
Establish a mapping between fine-grained features and established neuropsychological domains such as executive control, attention, and processing speed. This ensures that digital biomarkers remain clinically meaningful and interpretable to neurologists and psychologists, bridging the gap between computational signals and cognitive processes.
- **To develop predictive machine learning models for cognitive decline in PD**
Design and train machine learning models (e.g., Random Forest, XGBoost, Neural Networks) that utilize extracted trial-level features to detect Parkinson's-related cognitive decline. Emphasis will be placed on sensitivity

to early changes, as well as robustness across large mobile datasets such as mPower.

- **To integrate explainable ai for clinician-facing interpretability**
Apply explainable AI methods (e.g., SHAP, LIME) to identify which cognitive features most strongly influence prediction outcomes. Outputs will be presented in interpretable formats (feature attribution charts, task-level importance scores) to provide clinicians with transparent insights into why a given prediction was made.
- **To validate and benchmark the framework against existing approaches**
Conduct validation studies using publicly available mobile datasets (e.g., mPower) and newly collected app-based task data. Compare the predictive performance of trial-level features against traditional aggregate measures, and evaluate accuracy, sensitivity, specificity, and interpretability to demonstrate the framework's added value.
- **To integrate the cognitive module into the multi-modal mobile application**
Embed the cognitive biomarker analysis framework into the group's multi-modal Parkinson's detection mobile app. This integration will enable the fusion of cognitive signals with other biomarkers (speech, motion, drawing), providing a comprehensive, scalable, and clinically relevant early detection system.

3.METHODOLOGY

3.1 Overall System Methodology

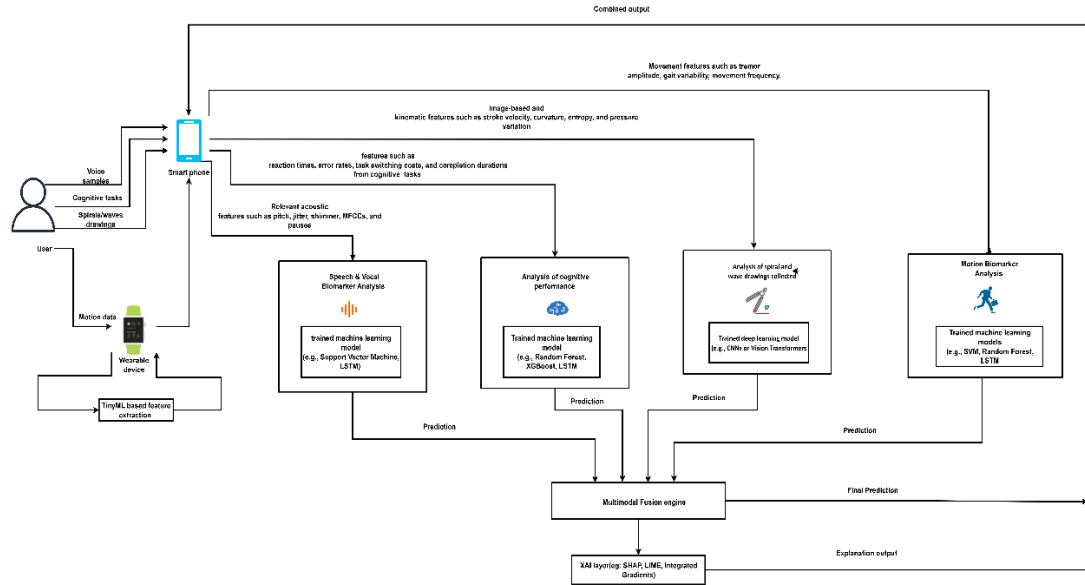


Figure 3.1 Overall system diagram

This project presents a multimodal, AI-driven diagnostic system for the early identification of neurodegenerative disease Parkinson's. Four main data streams are integrated into the solution: wearable sensor data, spiral and wave drawing analysis, speech and vocal biomarkers, and digital cognitive testing using tasks related to language, memory, and executive function. A dedicated module processes each input modality. To identify pauses, slurring, and monotony, speech signals are analyzed. Wearable sensors record tremors and abnormalities in movement. Image-based analysis is used to evaluate drawing tasks for motor abnormalities. Digital cognitive tests use task-based interaction data and natural language processing (NLP) techniques to evaluate verbal fluency, recall, attention, and problem-solving to identify early cognitive decline, the extracted features are combined in a Multi-Modal Prediction Engine that employs ensemble models and deep learning. TinyML-optimized lightweight models enable on-device deployment, guaranteeing real-time analysis while protecting user privacy. To improve trust and adoption, explainable AI (XAI)

tools like attention maps and SHAP are integrated to make model decisions understandable to patients and clinicians.

3.2 Component Methodology

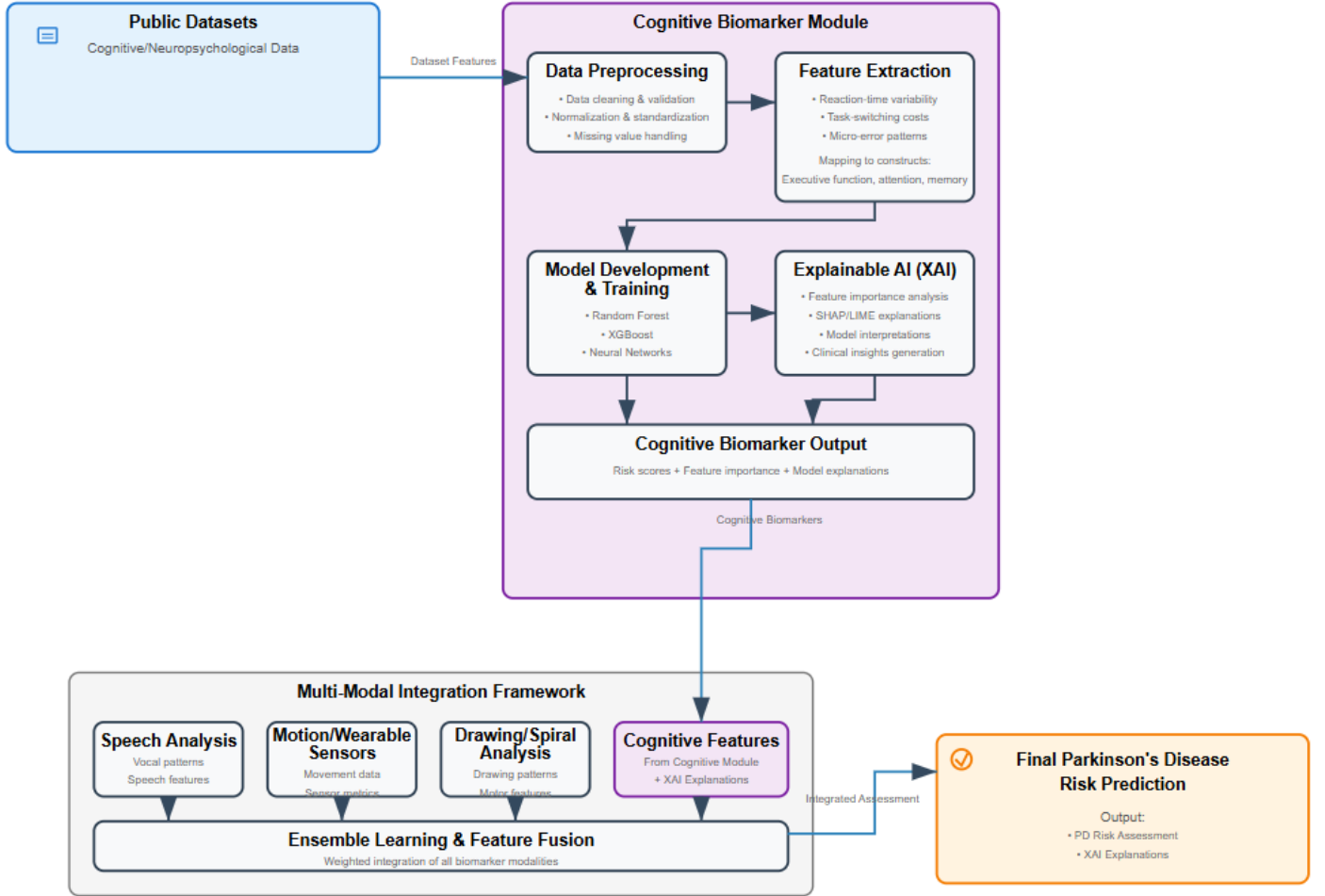


Figure 3.2 Component architecture

Dataset Selection and Preprocessing

This study will utilize publicly available Parkinson's disease datasets to train and validate the predictive models. Datasets such as mPower, PPMI, and other open-access repositories provide structured data from neuropsychological assessments and cognitive tasks, including memory recall, attention tests, reaction times, and executive function measurements. These datasets were selected due to their comprehensive coverage of cognitive performance in both Parkinson's patients and healthy controls,

allowing the extraction of features that are clinically relevant for early detection. The use of pre-existing datasets ensures that the study adheres to ethical standards and avoids the logistical and time-related constraints associated with primary data collection.

Preprocessing will be applied to ensure that the datasets are suitable for machine learning analysis. This includes handling missing values, normalizing numerical features, encoding categorical variables, and removing noise or inconsistent entries. In addition, data will be checked for class imbalance, and if necessary, resampling techniques such as SMOTE or under sampling will be employed to create balanced training and validation sets.

Feature Engineering

The next phase focuses on deriving cognitive and neuropsychological features from the preprocessed datasets. Features will be chosen based on established literature linking cognitive deficits to PD. For example, memory performance will be assessed through accuracy scores and error patterns in recall tasks, while executive function will be represented by reaction time variability, task-switching efficiency, and working memory span. Attention and processing speed will be captured through reaction time measures and accuracy under time-constrained tasks.

Feature engineering will also include aggregating trial-level data into higher-order constructs that align with clinical neuropsychological domains, enabling the model to capture subtle impairments that may not be evident from aggregate scores alone. The methodology allows flexibility to incorporate additional behavioral or clinical features, such as sleep quality or mood scores, if these become available or relevant for enhancing predictive power.

Model Development

Machine learning models will be developed to analyze the extracted features and predict Parkinson's disease status. The study will explore a range of algorithms, including deep neural networks, random forests, and support vector machines, to

identify the approach that best captures the patterns associated with cognitive impairment in PD.

Training will be conducted using supervised learning approaches, with Parkinson's diagnosis or early cognitive decline as the target labels. Model hyperparameters will be optimized using grid search or Bayesian optimization. Regularization techniques such as dropout and L2 penalties will be applied to prevent overfitting and improve generalizability. Model performance will be evaluated using cross-validation and metrics including accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (AUC-ROC).

Explainable AI (XAI) Integration

An essential component of this research is the incorporation of **Explainable AI (XAI)** techniques to ensure model transparency. Methods such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) will be applied to interpret the influence of individual features on model predictions. This allows identification of the most relevant cognitive and neuropsychological features contributing to PD detection, providing insights that are interpretable by clinicians. The integration of XAI ensures that the system is not only accurate but also ethically and clinically meaningful, fostering trust in the AI-based decision support tool.

Mobile Application Deployment

The final phase of the methodology involves integrating the trained predictive model and XAI framework into a mobile application, which serves as the user interface. The app will present cognitive and neuropsychological tasks that mirror those available in the datasets, such as memory recall, reaction-time challenges, and attention exercises. While the model is trained entirely on publicly available data, the mobile application provides a platform to deliver real-time predictions to users. The application will also display interpretability outputs, allowing users and clinicians to understand how performance in specific tasks contributes to the predicted PD risk. Data privacy, security, and ethical compliance will be integral to the application design.

Through this structured methodology, the study ensures a rigorous and systematic approach to leveraging existing datasets for predictive modeling, incorporating explainability, and providing a clinically relevant mobile interface for PD risk assessment. This framework maintains flexibility to incorporate additional cognitive or behavioral features as needed, ensuring the research remains adaptable and comprehensive.

3.3 Tools and Technologies

Table 3.1 Tools and technologies

Tool/Technology	Description	Purpose
Python	Versatile scripting language with libraries for data analysis.	ML model development, scripting, and prototyping.
Git	Distributed version control system.	Code collaboration, versioning, and repository management.
VS Code	Lightweight, extensible code editor.	Integrated development environment for coding and debugging.
Figma	Collaborative interface design tool.	Wireframing and prototyping the user interface.
Firebase	Backend-as-a-Service platform.	Secure data storage, authentication, and real-time analytics.
Scikit-learn	ML library for classical algorithms.	Model training (e.g., SVM) and evaluation metrics.
React Native	JavaScript framework for mobile application development	Mobile application development

4.PROJECT REQUIREMENTS

4.1 Functional requirements

- **User Management** –The system should allow users to register, log in, and manage their profiles securely.
- **Task Interaction** – The application should provide cognitive tasks and record user responses during interactions.
- **Data Processing** – The system should preprocess collected data and extract meaningful cognitive features.
- **Multimodal Integration** – The system should combine data from different sources to enhance prediction accuracy.
- **Predictive Model** – The system should apply a trained deep learning model to predict early cognitive decline.
- **Explainability** – The system should provide simple explanations of how predictions are made.
- **Result Visualization** – The system should display predictions and insights in a clear, user-friendly format.

4.2 Non-functional requirements

- **Usability**
The system should provide a simple, intuitive interface suitable for both patients and healthcare professionals.
- **Accuracy**
The diagnostic predictions must achieve a high degree of precision and recall ensuring reliability in early detection.
- **Performance**
The system should process inputs and generate results within an acceptable response time without delays.

- **Scalability**

The architecture must support large-scale data collection and training as the number of users grows.

- **Reliability**

The system should operate consistently without frequent failures, ensuring stable and continuous monitoring.

- **Security**

All patient data must be encrypted in storage and transmission, complying with medical data privacy standards.

4.3 User requirements

4.3.1 Patients

- Easy-to-use interface suitable for elderly and impaired users.
- Clear visual, audio, or voice-assisted instructions.
- Progress tracking with simple charts and reminders for regular use.
- Multilingual support: English, Sinhala, Tamil.

4.3.2 Healthcare Providers

- Access to detailed diagnostic reports with clear explanations.
- Integration with existing hospital/clinical systems.
- Tools to track patient progress over time.

4.3.3 Administrators

- User and access management features.
- System usage analytics and monitoring.
- Strong privacy and security compliance.

4.4 System requirements

4.4.1 Hardware

- Smartphone or tablet with sufficient memory and storage.
- Touchscreen support with good display quality.
- Microphone for voice input.
- Internet connectivity for updates and data synchronization.

4.4.2 Software

- Support for Android and iOS platforms.
- Machine learning frameworks for model deployment.
- Secure local and cloud database for data storage.

4.4.3 Security & Performance

- Encrypted and backed-up data storage.
- Role-based access with authentication.
- Low-latency performance and error monitoring.
- Modular design for easy updates.

4.5 Wireframes

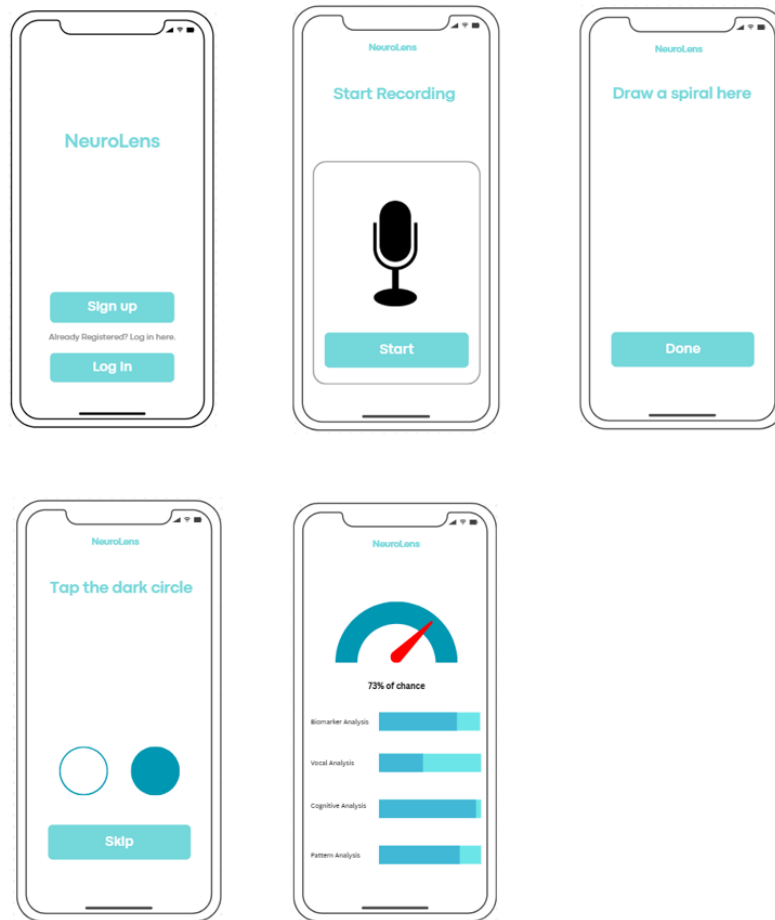


Figure 4.1 Wireframes

5. COMMERCIALIZATION

Free Tier

The free tier provides access to basic assessments and feedback using selected modules of the app. This tier is ideal for users who want an initial screening or personal monitoring tool. Limitations include restricted access to advanced analytics, multi-modal integration, and real-time AI-powered predictions. Wearable device integration and explainable AI (XAI) outputs are not available in this tier, ensuring the app remains lightweight and accessible while providing essential early detection features.

Paid Tier (LKR 1990)

The paid subscription unlocks the full potential of the app, including complete multi-modal interaction. **This tier includes wearable device integration**, enabling continuous, real-time monitoring of motor and cognitive signals. This makes the app far more proactive, moving beyond periodic self-assessments to continuous digital biomarker collection. In addition, premium users gain access to explainable AI (XAI) outputs, which highlight how different biomarkers contribute to predictions, along with AI-powered insights and early detection predictions. This tier is designed to provide comprehensive, clinically relevant tools for patients, caregivers, and clinicians.

This tiered approach gives users flexibility according to their needs, supports sustainable app development, and allows continuous improvement. Regular updates will add new features and enhancements, with premium offerings delivering advanced monitoring, interpretability, and integration capabilities.

6. WORK BREAKDOWN CHART

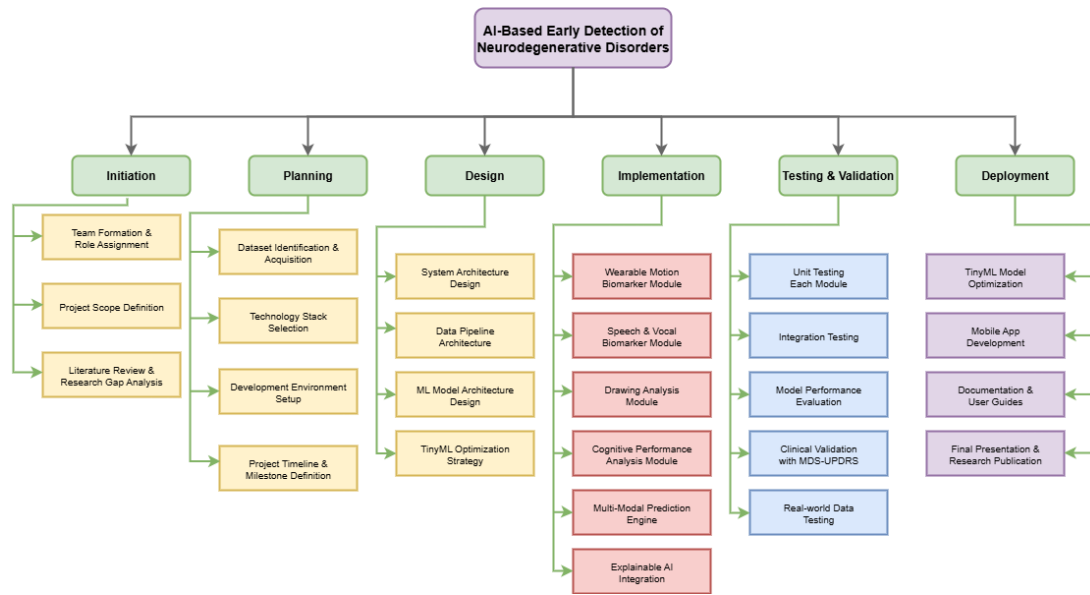


Figure 6.1 Work breakdown chart

7. GANTT CHART



Figure 7.0.1 Gantt Chart

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APPENDICES

Appendix A – Plagiarism Report

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